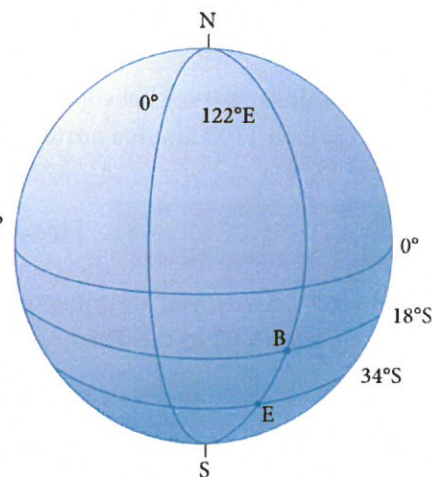


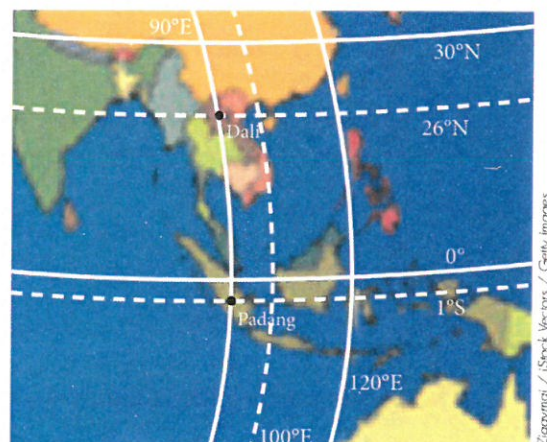


EXERCISE 15.02 Distances on the Earth's surface

- 1 **Example 3** Esperance (E) and Broome (B) are on the 122°E meridian of longitude. Esperance is at 34°S , 122°E and Broome is at 18°S , 122°E .
- Use a calculation to show that the change in the angle between Broome and Esperance is 16° .
 - Calculate the distance between Broome and Esperance, correct to the nearest 10 km.



- 2 Bernadette chartered a plane in Dali in China to fly to Padang in Sumatra. The plane flew along the 100°E meridian of longitude. Dali has position coordinates 26°N , 100°E and Padang's position is 1°S , 100°E .
- What is the angular difference between Dali and Padang along the 100°E meridian of longitude?
 - How far is it from Dali to Padang? Answer correct to the nearest 100 km.
 - The flight took 12 hours. Calculate the plane's average speed.



- 3 Buenos Aires in Argentina is at 34°S , 58°W and Bridgetown in Barbados is at 13°N , 58°W .
- Explain why we add 34° and 13° to determine the angle between Buenos Aires and Barbados along the 58°W meridian of longitude.
 - Calculate the distance from Bridgetown to Buenos Aires. Answer correct to the nearest 10 km.
- 4 Durban in South Africa is at 30°S , 30°E and Odessa in the Ukraine is at 45°N , 30°E .
- What is the change in the angle along the 30°E meridian of longitude between Durban and Odessa?
 - Calculate the distance from Durban to Odessa, correct to the nearest 100 km.
- 5 Port Fairy in Victoria and Weipa in Queensland are both on the 142°E meridian of longitude. Port Fairy is at 38°S and Weipa is 12°S . How far is it from Weipa to Port Fairy?

- 6 Jonathan was born in Port Hedland, 20°S, 119°E in the north of Western Australia and he works for the council in Sandstone 28°S, 119°E. On long weekends, he catches a light plane home to visit his family in Port Hedland.
 - a How far is it from Sandstone to Port Hedland?
 - b The light plane flies at an average speed of 120 km/h. How long does Jonathan's flight home take?
- 7 A mining survey helicopter left Perth, 32°S, 116°E. During an electrical storm, the plane crashed 1100 km due north of Perth. Determine the position coordinates of the crash site.

INVESTIGATION

How accurate is it?

In this activity, you are going to compare your answers to the distance calculations you made in Exercise 15.02 to the distances provided by online calculators.

- 1 Calculate the distance between the locations 32°S, 12°E and 20°S, 120°E.
- 2 Access the 'distance calculator', which is available on NelsonNet, and compare the distance to your calculated distance.

Great-circle distance between two points

Enter the co-ordinates into the text boxes to try out the calculations. A variety of formats are accepted, principally:

deg-min-sec suffixed with N/S/E/W (e.g. 40°44'55"N, 73 59 11W), or
signed decimal degrees without compass direction, where negative indicates west/south (e.g. 40.7486, -73.9864):

Point 1:	32S	,	12E	Distance:	10420 km (to 4 SF)
Point 2:	20S	,	120E	Initial bearing:	116°24'54"
				Final bearing:	053°55'32"
				Midpoint:	39°36'55"S, 070°02'09"E

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- 3 Compare your answers to the questions in Exercise 15.02 to those obtained using the website.
- 4 What conclusions can you make?

15.03 LONGITUDE AND TIME DIFFERENCES

In Chapter 1, we learnt about time zones, time differences and universal time coordinated; local time compared to the time in Greenwich near London, England.

We use time zones for convenience. After all, it would be very confusing to have the time on the eastern side of a city different from the time on the western side. A location's 'true time' is determined by its longitude and all locations on the same longitude have the same true time.

IMPORTANT

A 15° change in longitude corresponds to a 1 hour time difference.